Speech-model-v3-TTS

Fine-tuning ThonburianTTS (F5-TTS) for natural Thai speech, masculine / feminine voice profiles and gap-aware synthesis ("…")

TTS-Lab · result report v1 · 2026-11-01 · every number in this document is read from the files produced by the code in TTS-Lab/ (scores.csv, manifest.csv, meta.parquet, log.json, valcmp*.json); nothing is estimated unless explicitly labelled "estimate".

Abstract

We fine-tune the pretrained ThonburianTTS model (F5-TTS, 337 M parameters, flow matching, 24 kHz) on a speaker-diverse, quality-filtered subset of Chalermdej/yodas2_sidon_th_tts (12.4 h, 10,697 S+ clips, 2,625 speaker ids) in a single 45-minute run on one Vertex AI Spot A100 (estimated USD 1.32; USD 1.59 for the whole project including four failed start-up attempts). Training adds 20 % synthetic "gap" utterances (A + [soft murmur or pause] + B, A + gap, gap + B) paired with the marker "…", which already exists in the vocabulary, so that an unreadable sign can be voiced as a natural pause or quiet hum instead of a guessed word. Voice selection (masculine / feminine) is carried by the reference-audio prompt as in F5-TTS; four reference voices (two per presentation) come from held-out speakers. On held-out speakers the flow-matching loss on real clips falls from 1.81 (baseline) to 0.56 (validation) and from 1.77 to 0.56 (test). On a fixed 110-clip synthesis set scored with an independent ASR (Whisper large-v3-turbo) and an independent speaker-verification model (WavLM-base-plus-sv), speaker similarity to the reference voice rises (0.911 → 0.932), the fine-tuned model produces audible silences or hums at the "…" marker where the baseline mostly speaks through it, while the median character error rate is essentially unchanged (0.022 vs 0.020). No human MOS/CMOS was collected; weak evidence is flagged throughout.

1 Goals and research questions

The goal is a Thai TTS component for a sign-language application: natural speech, a user-selectable masculine or feminine voice, and graceful handling of signs the recogniser could not read ("ฉัน ___ แม่", "ฉันอยากไปประเทศ ___"), where the voice should hesitate softly rather than say a wrong word. **RQ1** does fine-tuning on quality-filtered Thai speech improve the baseline? **RQ2** can masculine / feminine voice profiles be selected consistently? **RQ3** can a gap marker be taught without changing the architecture? **RQ4** can this be done within a hard cloud budget (target ≤ USD 3–4; user cap for everything USD 5–6)?

Scope: emotion / style conditioning was **not** implemented (no reliable labels in the data). Speaking-rate control is inherited from F5-TTS (duration scaling). No ablation runs were made (one planned run), see Section 9.

2 Base model: ThonburianTTS

ThonburianTTS (biodatlab/ThonburianTTS, checkpoint megaF5/mega_f5_last.safetensors, snapshot ac854d8) is an F5-TTS model: a Diffusion Transformer (DiT) trained with conditional flow matching to infill a masked span of a 100-channel mel spectrogram (24 kHz, hop 256, Vocos mel) given character-level text and the unmasked mel context (the reference audio at inference). Architecture, taken from the baseline loader and verified by loading all checkpoint keys without mismatch: dim 1024, depth 22, 16 heads, FF multiplier 2, text embedding dim 512 with 4 ConvNeXt text blocks, vocabulary 2,586 characters (+1 filler), 337 M parameters. Inference: Euler ODE, 32 function evaluations, classifier-free guidance 2.0, Vocos vocoder. The total output duration is fixed before sampling (reference duration + estimated text duration), which is why the marker needs special duration handling (Section 5.4).

Fine-tuning compatibility (checked in code): weights load into the f5-tts 1.1.22 DiT/CFM training model with all keys matching; the CFM forward returns the flow-matching loss on a random masked span (mask fraction 0.7–1.0, audio-condition drop 0.3, text-condition drop 0.2; F5 defaults). The model has no speaker embedding (identity comes from the audio prompt) and no emotion/style conditioning. LoRA was not used: full fine-tuning fit on one 40 GB A100 (peak 31.8 GB).

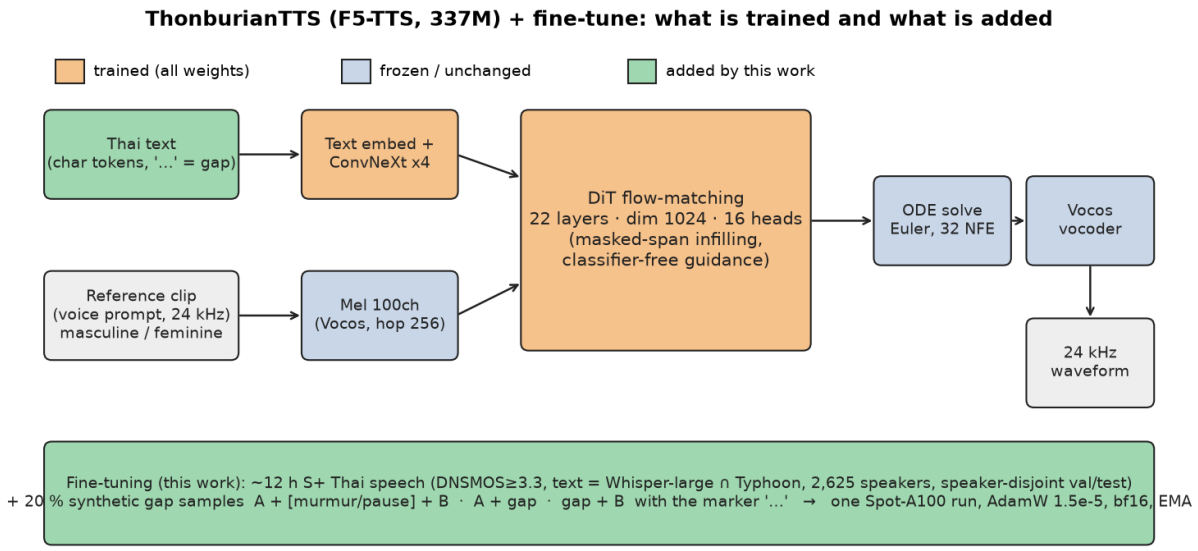


Figure 1. ThonburianTTS / F5-TTS pipeline and what this work adds (green).

3 Dataset

Source: Chalermdej/yodas2_sidon_th_tts (Hugging Face, parquet, 59 files, 26.7 GB, licence tag cc-by-3.0): YODAS2 Thai audio enhanced with Sidon, graded S+/S/A+, 24 kHz. Fields: audio, utt_id, text, speaker_id, grade_avg, duration, dnsmos_overall/signal/background, original_ref, whisper_large, whisper_typhoon, typhoon_audio. The non-audio columns of all 56 shards (54 train + 2 test) were read with parquet range requests (no audio downloaded) to profile the whole dataset.

Grade	Clips	Hours
S+	78,180	82.36
S	51,518	60.73
A+	12,229	12.94
Total (56 shards)	141,927	156.04

Property	Value
Distinct speaker_id / distinct video ids (utt_id prefix)	4,199 / 4,161
Clip duration (s): min / p25 / median / mean / p75 / max	1.50 / 2.58 / 3.46 / 3.96 / 4.76 / 17.51
DNSMOS overall (all grades): mean ± sd (min–max)	3.33 ± 0.14 (3.00–3.77)
Transcript length (characters): median	43

Caveat on speaker identity. speaker_id is an anonymous number and utt_id begins with a YouTube video id; neither is a verified person. The counts above (more speaker ids than video ids) show the ids are not one-to-one with videos. We treat speaker_id as a speaker group for splitting and per-speaker caps, never as ground truth, and we do not expose video ids. The dataset has no gender label; none was inferred from ids or metadata (Section 4.4).

4 Data preparation

Implemented in TTS-Lab/prep.py (CPU, local).

4.1 Selection of clips

(1) S+ clips from train shards only. (2) 14 of the 54 train shards, evenly spaced (indices 0, 4, 8, … 53), to avoid downloading 26.7 GB (≈ 6.9 GB downloaded; raw shards deleted afterwards). (3) duration 2–12 s. (4) DNSMOS overall ≥ 3.3. (5) **Transcript agreement**: character-level difflib similarity (spaces and punctuation removed) between the dataset text and both the whisper_large and whisper_typhoon columns ≥ 0.9 (mean agreement over a random S+ sample 0.977, median 1.0). (6) at most 30 clips per speaker id (seeded random), so that few speakers cannot dominate. (7) audio with peak ≥ 0.999 (clipping) or < 0.05 (nearly silent) is dropped (0 clips removed by this rule). Audio is stored as 24 kHz mono 16-bit FLAC; no loudness normalisation, VAD trimming or

re-segmentation, so natural pauses and dynamics are preserved.

On a random sample of 3,000 S+ train-shard clips, 53.4 % pass steps (3)–(5); extrapolated, the whole S+ pool would give about 49.6 h before the speaker cap (estimate). S and A+ were **not** used (quality over volume; the 12 h subset already trained for ≈ 20 epochs).

Split	Clips	Hours	Speaker ids
train	10,173	11.80	2,505
val	257	0.30	61
test	267	0.31	59
Total	10,697	12.41	2,625

4.2 Speaker-disjoint splits

By speaker id: md5(seed + id) mod 100 < 2 → test, < 4 → val, else train (≈ 2 % / 2 % / 96 %). No speaker id appears in two splits, so validation and test measure generalisation to unseen ids (ids are not verified identities, so residual leakage of the same person is possible). Test transcripts were not de-duplicated against training text.

4.3 Gap augmentation ("…")

A gap sample is built on the fly from clips of the **same speaker id**: **mid** (60 %) = A + gap + B with text "A … B" (A+B ≤ 15 s); **end** (25 %) = A + gap, text "A …"; **start** (15 %) = gap + A, text "… A". The number of gap items is 20 % of the real train items (1,989 realised in train, 51 in val). Gap length is uniform 0.45–1.4 s (mid/end) or 0.35–0.9 s (start). Gap audio: (a) room tone from the three quietest 80 ms windows of A, tiled with 10 ms cross-fades (Sidon-enhanced clips are very clean, so this is nearly silent); (b) with probability 0.55 a soft hum is added inside the pause: the loudest stretch (0.25–0.9 s) of A, low-passed at 500 Hz (4th-order Butterworth), scaled to 30 % of A's RMS, Hann-windowed. Segments are joined with 12 ms cross-fades. "…" is already in the pretrained vocabulary, so the architecture is unchanged. **Limitation:** the hum is signal processing, not recorded human murmuring; no standalone filler utterances were identified in the dataset to use instead.

4.4 Voice presentation labels (masculine / feminine)

Per speaker id, the median YIN F0 over voiced frames (frame energy > 0.5 × median; 60–400 Hz) is computed: ≥ 180 Hz → "feminine", ≤ 150 Hz → "masculine", otherwise "unk" (excluded from voice profiles). Over 2,625 speaker ids: feminine 1,200, masculine 1,112, unk 313. This is a **preliminary acoustic signal only** (voice presentation, not gender identity) and was not verified by listening to a sample. The labels are used only to choose reference voices; they are not a training input.

4.5 Reference voices (voice profiles)

Profile	Presentation	Speaker median F0 (Hz)	DNSMOS
masculine_1	masculine	139.5	3.72
masculine_2	masculine	114.6	3.57
feminine_1	feminine	216.4	3.54
feminine_2	feminine	201.8	3.6

For each presentation the two test-split speakers with the highest mean DNSMOS were taken, and for each the 5–9 s clip with the highest DNSMOS; the transcript is the dataset text. The references come from test speakers never seen in training, so cloning from unseen prompts is what is evaluated.

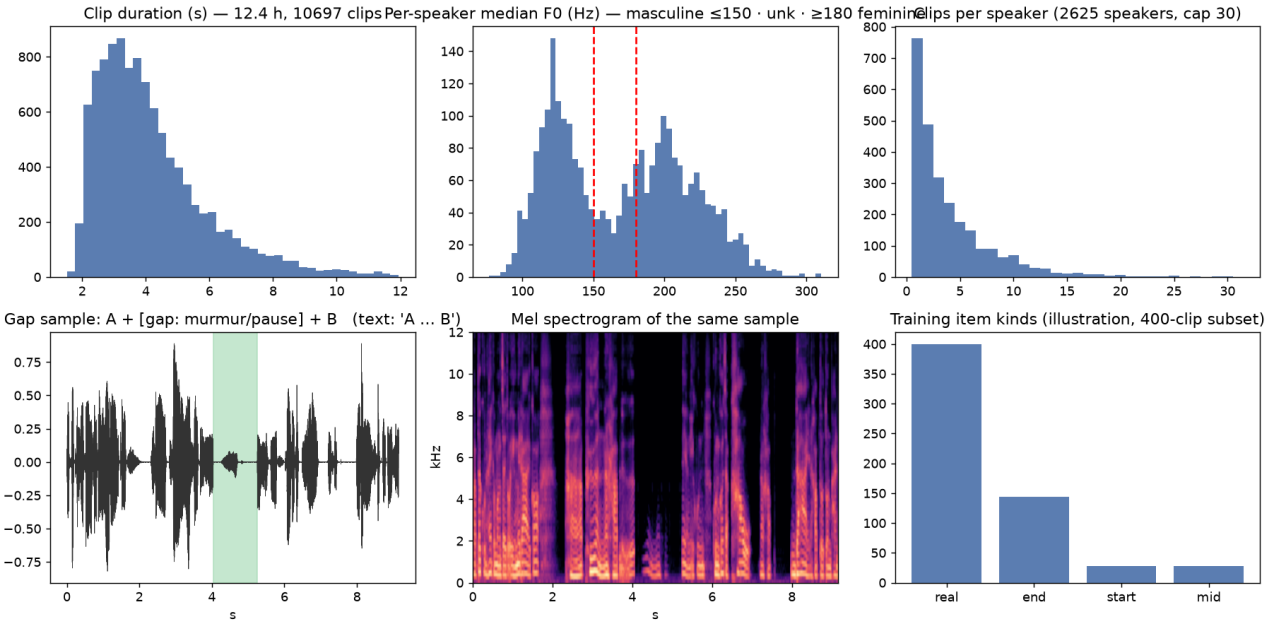


Figure 2. Training-subset statistics and an example gap sample (waveform + mel).

5 Fine-tuning

5.1 Strategy and settings

Full fine-tuning of all 337 M parameters from the pretrained EMA weights, one planned run, no architecture change and no new parameters. LoRA and added speaker/emotion embeddings were rejected: the model already clones voices from the audio prompt, and new embeddings would not start from pretrained knowledge and would need more data than the budget allowed. Forgetting was checked through held-out loss and ASR (Section 7), not assumed away.

Setting	Value
Framework	f5-tts 1.1.22 model code (DiT, CFM), own training loop (train.py)
Initialisation	EMA weights of mega_f5_last.safetensors (all keys matched)
Train items	12,162 = 10,173 real + 1,989 gap, 15.8 h; validation items 238 (200 real + 38 gap)
Batching	dynamic by frames: 19,200 mel frames per micro-batch × 2 accumulation = 38,400 frames (≈ 410 s of audio) per update; length-bucketed; OOM fallback ×0.75 (never triggered)
Optimiser	AdamW, peak LR 1.5e-5, betas (0.9, 0.999), weight decay 0.01, gradient clip 1.0, fused
Schedule	linear warm-up 100 updates, cosine decay to 10 % of peak; progress = max(elapsed / 45 min, epoch / 30) (time-budgeted)
Precision	bf16 autocast, fp32 master weights
EMA	decay = min(0.995, (1+t)/(10+t)) per update
Validation / selection	EMA model, flow loss on validation items every 150 updates, fixed seed; best kept; patience 5 (never triggered)
Run length	2,774 updates in 45.1 min (≈ 0.98 s/update), 20th epoch partial; best validation loss 0.5990 (final evaluation)
Hardware	1 × NVIDIA A100-SXM4-40GB (Vertex AI custom job, Spot, asia-southeast1); peak GPU memory 31.8 GB; container pytorch-gpu 2.4 (py3.10), torch 2.4.0+cu124
Seeds	training seed 0; gap recipes seed 0 (train) / 1 (val); split hash seed 0

5.2 Training curve

The EMA validation flow-matching loss decreased monotonically from 0.672 (update 150) to 0.599 (update 2774) with a flattening tail (Figure 3, left); validation never rose, so no overfitting signal appeared in this window. The loss is stochastic (random time and mask), hence the fixed validation seed.

5.3 Cloud-compute plan and cost

Cloud GPU was used for one step only (training + generation of fine-tuned audio). Data selection, preprocessing, baseline audio, scoring, figures and the notebook ran locally (RTX 2050 4 GB, 16 GB RAM), because a 337 M-parameter fp32 model with Adam needs more than 4 GB. Spot A100 was chosen because the project has no L4 quota and it is the cheapest available option that fits 31.8 GB. Three regions were raced (first to get a GPU wins; the others are cancelled while pending, which is unbilled). Cost = run minutes × USD 1.55/h assumed price (an estimate, not an invoice).

Attempt	Outcome	Run (min)	Est. cost (USD)
1	failed: torchvision / torch ABI mismatch after pip install	0.8	0.021
2	failed: torchaudio CUDA-13 wheel on a CUDA-12.4 container	2.3	0.061
3	failed: same torchaudio issue (version constraint ignored)	2.3	0.058
4	environment OK, baseline audio generated; training crashed (invalid hum-length range for gaps < 0.45 s) - fixed and all 2,040 gap recipes re-tested locally	4.9	0.126
5	succeeded: training + fine-tuned audio generation	51.1	1.319
Total		61.4	1.585

After the run the Cloud Storage bucket was emptied (0 objects), all 15 Vertex custom jobs were deleted (0 remaining), and no VMs, disks, addresses or other buckets existed. Bucket storage of ≈ 1.2 GB for about two hours is negligible and not included; LLM-agent usage is not part of this figure.

5.4 Inference with gaps

F5-TTS fixes the output duration before sampling: reference duration + (UTF-8 bytes of the generated text) × (reference seconds per byte) / speed. The marker "…" would get only a few bytes of time, so tts.py instead uses duration = bytes(text without "…") × rate / speed + n_gaps × gap_seconds (default 0.8 s, user-adjustable), with a minimum of 0.9 s of generated speech. The minimum was added after a first scoring round showed that single-word clips (≈ 0.47 s) were unusable for both models and for Whisper; words were regenerated for both models with the same rule. Text normalisation (identical for training and inference): Thai digits → Arabic; "[?]", "___", "..", "…" → one "…" surrounded by spaces; whitespace collapsed; at most 400 characters; longer text is split at sentence boundaries into segments of ≤ 13 s. Baseline and fine-tuned models are always called through the same code with the same prompts.

6 Evaluation protocol

Fixed synthesis set, 110 clips per model: 7 full sentences × 4 voices (28); 3 gap sentences × 4 voices (12, one contains two gaps); 5 word groups × 5 words × 2 voices (masculine_1, feminine_1) = 50; and 10 transcripts from test-split speakers × 2 voices = 20. Same texts, references, seed and settings for both models (NFE 32, CFG 2.0, speed 1.0, gap 0.8 s). Baseline = original ThonburianTTS checkpoint; fine-tuned = this work (EMA best checkpoint, fp16). Texts are listed in Appendix A.

Metrics (automatic; TTS-Lab/evaluate.py):

• **CER**: Whisper large-v3-turbo (openai/whisper-large-v3-turbo, language th, CPU fp32; independent of the TTS) transcribes each clip; character-level Levenshtein distance after removing spaces and punctuation, divided by reference length (gap markers removed from the reference). Whisper can loop (the baseline clip s01_masculine_1 was transcribed as "สวัสวัสวัส…", CER 2.05), so the headline is the **mean of CER clipped at 1** and the **median** is reported as the robust statistic. No Thai word segmentation was used: these are character, not word, error rates.

• **Speaker similarity**: cosine similarity of WavLM-base-plus-sv x-vector embeddings (16 kHz; clips under 1 s zero-padded) between the generated clip and the reference prompt of that voice. It is embedding similarity, not identity, and not a gender or pitch measure.

• **F0 spread**: standard deviation (Hz) of YIN F0 (60–400 Hz) over all frames; a crude proxy for pitch variation (includes unvoiced frames, so only the baseline-vs-fine-tuned difference is interpretable).

• **Pause span**: longest low-energy stretch strictly inside the utterance (20 ms RMS frames below 0.15 × the 90th-percentile RMS); a hum does not count as a pause. Energy plots (Figure 4) are the primary gap evidence.

• **Held-out flow-matching loss** for pretrained and fine-tuned models with the same loader, seed and batching (4,000 frames) on the speaker-disjoint validation (257 clips) and test (267 clips) splits. The baseline was trained on other data, so this partly measures domain adaptation to clean, Sidon-enhanced speech.

• **RTF** = generation time / audio duration, on the A100 for sentence, gap and test clips (single clip, warm, fp32 weights, including the vocoder).

• **Not measured**: MOS, CMOS, MUSHRA (no listening test), MCD (needs aligned references), emotion accuracy, speaker-verification EER, long-form stability. No claim about perceived naturalness is made.

7 Results

7.1 Held-out flow-matching loss

Set	Baseline	Fine-tuned
Validation, real clips only	1.814	0.557
Test, real clips only	1.774	0.556
Validation incl. gap items	1.704	0.552
Test incl. gap items	1.765	0.590

The large drop on real clips mainly reflects adaptation to this recording domain; by itself it is not evidence of better perceived quality.

7.2 Synthesis metrics by prompt type

Type	Model	n	CER (clip 1)	CER median	Spk-sim	F0 std (Hz)	Pause span (s)
Sentences	baseline	28	0.088	0.022	0.918	56.9	0.222
	fine-tuned	28	0.047	0.023	0.943	69.0	0.311
Test-split sentences	baseline	20	0.096	0.028	0.919	56.0	0.257
	fine-tuned	20	0.057	0.000	0.943	70.6	0.303
Gap sentences	baseline	12	0.160	0.000	0.883	71.0	0.273
	fine-tuned	12	0.095	0.000	0.888	90.4	0.423
Single words	baseline	50	0.311	0.125	0.756	78.3	0.021
	fine-tuned	50	0.294	0.062	0.760	102.6	0.015
All non-word clips	baseline	60	0.105	0.022	0.911	59.5	0.244
	fine-tuned	60	0.060	0.020	0.932	73.8	0.331

Paired bootstrap over the 60 non-word clips (5,000 resamples; mean of fine-tuned − baseline, 95 % interval): speaker similarity +0.021 [+0.012, +0.028] (fine-tuned higher on 47, lower on 13, equal on 0 clips); clipped CER -0.045 [-0.090, -0.009] (lower on 22, higher on 13, equal on 25); F0 std +14.4 Hz [+10.1, +18.8]. These intervals describe these 60 clips (10 sentences × 4 voices + 10 test sentences × 2 voices); the clips are not independent draws from Thai text, so they are descriptive, not inferential.

Single words (50 clips, 0.89 s each after the minimum-duration fix): median CER 0.125 (baseline) vs 0.062 (fine-tuned); share of words with CER ≤ 0.34: 0.60 vs 0.66. Isolated words are hard for both systems and for Whisper (short-clip hallucination); similarity on such short clips is lower and noisier.

7.3 Per-voice speaker similarity (non-word clips)

Voice	Baseline	Fine-tuned	Δ
feminine_1	0.941	0.958	+0.016
feminine_2	0.912	0.921	+0.009
masculine_1	0.869	0.906	+0.037
masculine_2	0.933	0.942	+0.009

The improvement is not uniform across voices (see Δ). The reference voices span median F0 115–216 Hz; no listening test was run to confirm they are perceived as distinct masculine / feminine voices.

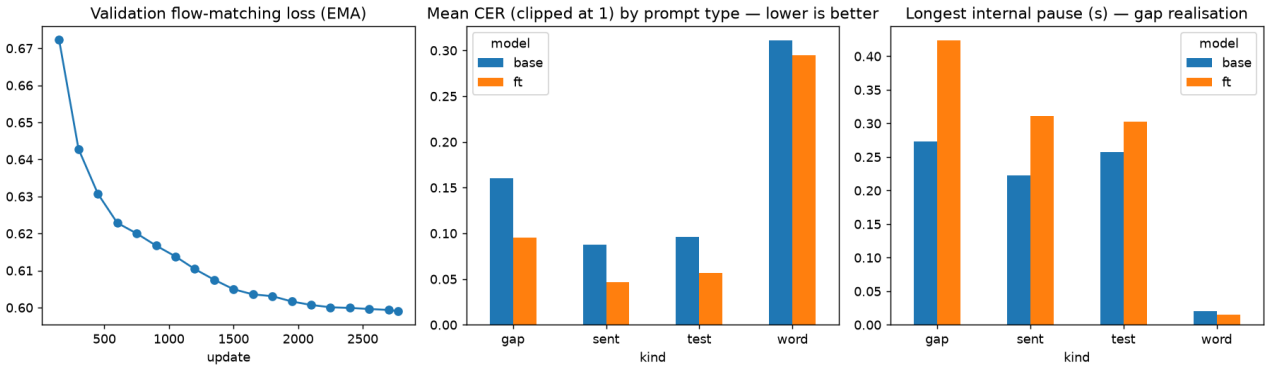


Figure 3. Left: validation loss during fine-tuning. Middle: mean CER (clipped at 1) by prompt type. Right: longest internal pause.

7.4 Gap realisation

The pause-span statistic rises for gap sentences (0.27 s → 0.42 s) but is a weak instrument because a hum does not count as a pause. Figure 4 shows the 50 ms energy of four generated gap sentences. For "ฉัน … แม่" the baseline speaks through the marker (feminine_1) or fills it with continuous low-level sound (masculine_1); the fine-tuned model inserts a soft hum followed by a short ≈ 0.1 s dip (feminine_1) or a ≈ 0.35 s stretch near −60 dB (masculine_1). For "ฉันอยากไปประเทศ …" the fine-tuned model stops speaking ≈ 0.5 s before the end of the clip, leaving a silent tail, whereas the baseline speaks until close to the end. For the two-gap sentence the fine-tuned model produces silences of roughly 0.5 s and 1.3 s at the two marker positions (and a ≈ 0.7 s leading silence), whereas the baseline babbles continuously at about −40 dB. In the ASR transcripts of "ฉัน … แม่" (4 voices) the baseline contains an inserted syllable in 3 of 4 clips ("ฉันถุ่นแม่", "ฉันถมแม่", "ฉันหมดแม่") and the fine-tuned model in 2 of 4 ("ฉันอุแม่", "ฉันหรุแม่"); the hum can be heard by Whisper as a sound. These are observations from signal plots and ASR on 12 clips per model, not perceptual ratings.

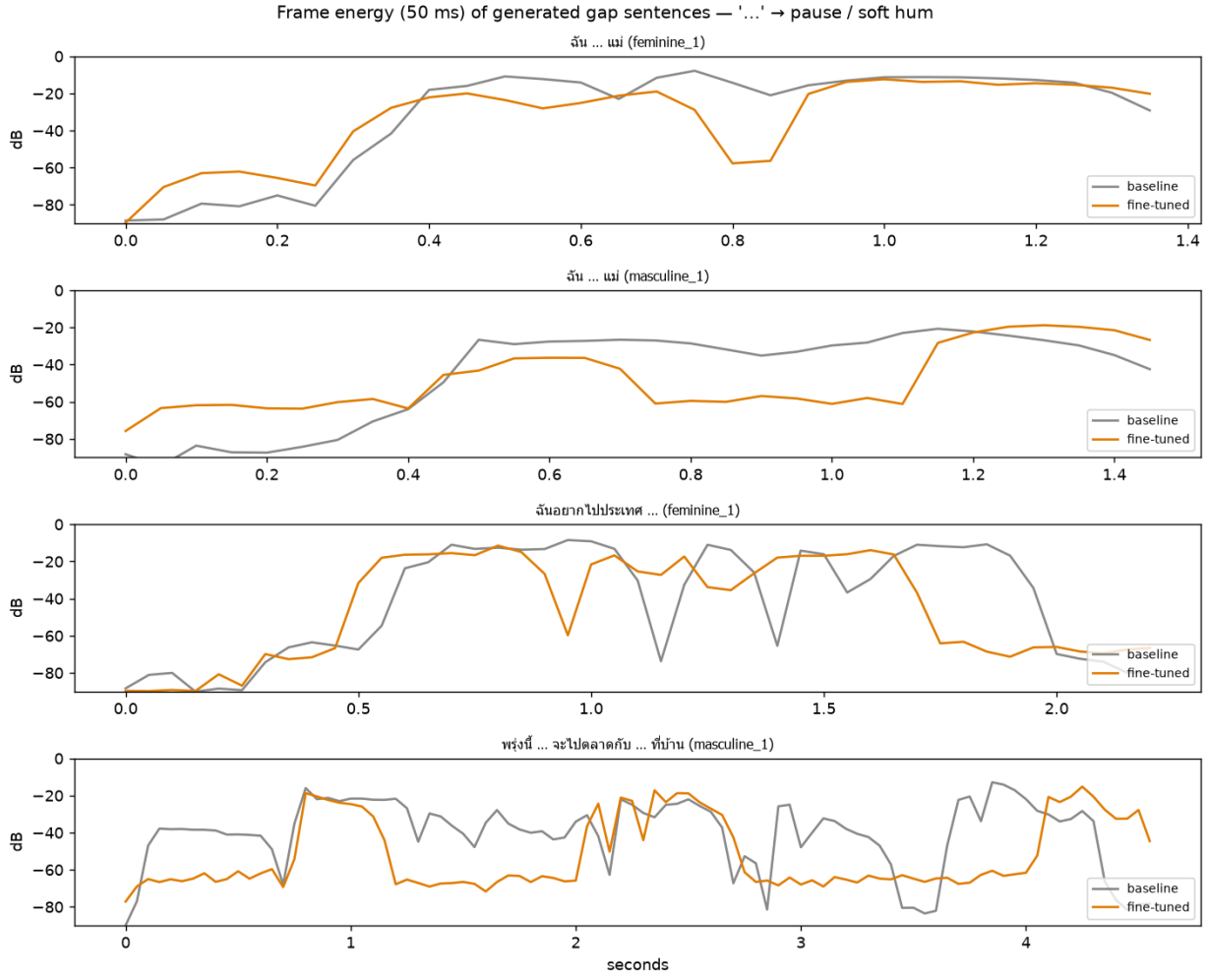


Figure 4. Frame energy of generated gap sentences, baseline (grey) vs fine-tuned (orange).

7.5 Speed and memory

RTF on the A100 (sentence, gap and test clips): baseline 0.282, fine-tuned 0.285 (identical architecture). Model loading ≈ 12 s on the A100. The package is 645 MB (fp16 weights). On the local RTX 2050 4 GB (shared with other processes) a short sentence was generated in ≈ 4 s through the HTTP API.

8 Discussion

RQ1 (improvement). Supported: much lower held-out flow loss; higher speaker similarity (+0.021); larger F0 spread (+14.4 Hz std); lower mean clipped CER (0.105 → 0.060). Not supported: the median CER is essentially unchanged (0.022 vs 0.020), so better pronunciation cannot be claimed; the mean difference comes from a few baseline clips that Whisper mis-transcribes. Perceived naturalness was not measured. **RQ2 (voices).** Four reference voices (2 + 2) give similarity 0.91–0.96 to their own prompts; selection is real (different prompts give different voices) but it is a property of prompting, not a learned speaker table, and the masculine / feminine labelling is unverified acoustic classification. **RQ3 (gaps).** Supported by energy plots and transcripts; the training signal is an approximation of natural hesitation. **RQ4 (budget).** Met: USD 1.59 estimated cloud cost for the whole project including failed attempts; the successful job lasted 51 minutes.

9 Limitations and threats to validity

• No human evaluation (MOS/CMOS/MUSHRA): no claim about perceived naturalness or expressiveness; F0 spread is a proxy only.

• Small, hand-written test set (110 clips; 7 + 3 + 25 + 10 texts; 4 voices): intervals describe these clips only; word- and gap-level results rest on 12–50 clips.

• Whisper large-v3-turbo has its own errors and hallucinations on short clips; CER was clipped and medians reported. The dataset's transcript columns also come from Whisper-family ASR, so agreement filtering and CER evaluation are not fully independent.

• Speaker ids are not verified identities (possible same-person leakage across splits); voice presentation is an F0 threshold, not verified; only four reference voices.

• Gap/hum training audio is synthetic DSP from clean, enhanced speech; real hesitations differ. Behaviour was verified with energy plots and ASR, not by listening.

• One run, no ablations: the effects of (a) data fine-tuning, (b) gap augmentation and (c) the inference-time duration rule cannot be separated. The baseline receives "…" with the same gap-aware duration, so the comparison is fair but does not show what a naive baseline call would do.

• The validation-loss comparison is confounded by domain shift (the pretrained checkpoint was trained on other data).

• No emotion / style control. Speaking-rate and gap-length controls were only spot-checked (not evaluated); long text (several segments) was not evaluated.

• Cost figures are estimates (run time × assumed Spot price). Local RTF on the RTX 2050 was not measured systematically.

• Licences: the ThonburianTTS repository ships MIT and CC-BY-NC-SA licence files; the fine-tuned weights derive from the pretrained weights and should be treated as non-commercial / share-alike until clarified with the authors. The dataset is tagged cc-by-3.0 (attribute Chalermdej/yodas2_sidon_th_tts, YODAS2, Sidon). No legal conclusion is drawn.

10 Reproducibility

Code (TTS-Lab/): prep.py (selection, FLAC export, manifest), train.py (gap synthesis, training loop), tts.py (inference), evaluate.py (refs / gen / score), vertex_tts.py (upload / launch / cleanup), make_figures.py, TTS-lab.ipynb (demo). In the application the model is served by the root docker-compose service tts (webapp/tts: GPU, male / female voice, threshold-aware speech, MMS CPU fallback). Model package: models/Speech-model-v3-TTS/ (model.safetensors, vocab.txt, voices.json, refs/, config.json, tts.py). Local environment: conda 490-TTS, Python 3.12.16, torch 2.11.0+cu128, f5-tts 1.1.22. Cloud: torch 2.4.0+cu124,

torchaudio 2.4.0, f5-tts 1.1.22 (torchvision removed).

Pipeline: python prep.py → python evaluate.py refs → python vertex_tts.py upload → python vertex_tts.py launch --minutes 45 → python evaluate.py score --dirs base=data/gen_base ft=data/gen_ft → python vertex_tts.py cleanup. Use: from tts import Speech; Speech("models/Speech-model-v3-TTS").say("ฉัน … แม่", voice="feminine_1"). Raw audio is not redistributed; data/manifest.csv lists sample id, speaker id, text, duration, DNSMOS, F0, voice presentation and split.

Appendix A Fixed evaluation texts

Sentences (7)

- สวัสดีครับ วันนีอากาศดีมาก ไปเดินเล่นทีสวนสาธารณะกันไหม
- ฉันอยากไปโรงพยาบาลเพราะรู้สึกไม่ค่อยสบาย
- คุณแม่ทำอาหารอร่อยมากเลยค่ะ ฉันชอบกินข้าวผัดทีสุด
- พรุ่งนีตอนเช้าเจ็ดโมงครึง เราจะนังรถไฟไปเชียงใหม่
- ขอบคุณมากนะคะ ทีช่วยสอนภาษามือให้ฉัน
- ทำไมวันนีถึงมาสายล่ะ มีอะไรเกิดขึนหรือเปล่า
- เด็กๆ ดีใจมากเมือได้รับของขวัญวันเกิดจากคุณพ่อ

Gap sentences (3)

- ฉัน … แม่
- ฉันอยากไปประเทศ …
- พรุ่งนี … จะไปตลาดกับ … ทีบ้าน

Word groups (5 × 5)

- group 1: พ่อ แม่ พี น้อง ลูก
- group 2: ข้าว น้ำ ขนม ผลไม้ ก๋วยเตียว
- group 3: บ้าน โรงเรียน โรงพยาบาล ตลาด ประเทศ
- group 4: กิน นอน เดิน ซือ เรียน
- group 5: ดีใจ เสียใจ เหนือย หิว รัก

Test-split transcripts (10, held-out speakers, random_state 0)

- หรือการสอบถาม อันนีก็ต้องเลือกวิธีการประเมินผลให้สอดคล้องกับงานแต่ละงาน
- จะเกิดขึนในช่วงหลายปีทีผ่านมา
- เมือนำไปใช้ร่วมกับสีอืน ๆ ซึงมีความเข้มมากกว่า จะช่วยเสริมความเข้มนันให้ดูเข้มข้นขึนกว่าปกติ
- ในการรวมคอลัมน์นะครับ เราก็ไปทีคอลัมน์ทีเราต้องการรวมนะครับ
- นอกจากนี ผู้สอนควรเตรียมตัวในการใช้สือ
- โดยดูผลจากการทีผู้เรียนสามารถบรรลุวัตถุประสงค์ตามทีได้กำหนดไว้
- จึงจะสามารถเข้าใจความหมายได้ชัดเจนยิงขึน
- และยังเปิดโอกาสให้ผู้เรียน แสดงผลงานของตนสู่สายตาผู้อืน
- และเป็นการแสดงสถานภาพทางสังคม
- การวางแผน การเตรียมการผลิตสือเสียง

References

- Chen et al., "F5-TTS: A Fairytaler that Fakes Fluent and Faithful Speech with Flow Matching", 2024 (arXiv:2410.06885).
- biodatlab, ThonburianTTS, github.com/biodatlab/thonburian-tts and huggingface.co/biodatlab/ThonburianTTS (IEEE iSAI-NLP 2025, doi:10.1109/iSAI-NLP66160.2025.11320472).
- Chalermdej, yodas2_sidon_th_tts dataset, huggingface.co/datasets/Chalermdej/yodas2_sidon_th_tts.
- Li et al., "YODAS: YouTube-Oriented Dataset for Audio and Speech", 2023.
- Siuzdak, "Vocos: Closing the gap between time-domain and Fourier-based neural vocoders for high-quality audio synthesis", 2023.
- Radford et al., "Robust Speech Recognition via Large-Scale Weak Supervision" (Whisper), 2022; openai/whisper-large-v3-turbo.
- Chen et al., "WavLM: Large-Scale Self-Supervised Pre-Training for Full Stack Speech Processing", 2022; microsoft/wavlm-base-plus-sv.
- Reddy et al., "DNSMOS P.835: A non-intrusive perceptual objective speech quality metric", 2022.
- de Cheveigné and Kawahara, "YIN, a fundamental frequency estimator for speech and music", JASA 2002.
- Lipman et al., "Flow Matching for Generative Modeling", 2023.